

Retail Return Risk Analytics & ML Report

E-Commerce Predictive Modeling & Risk Management | Machine Learning Pipeline

1. Executive Summary

Product returns create significant operational friction in e-commerce, incurring high reverse logistics costs, restocking delays, and inventory depreciation. This project provides an end-to-end predictive machine learning pipeline to classify high-risk retail orders prior to dispatch, enabling proactive operational interventions.

Key Objective: Predict whether an order will be returned (**target = 1**) or retained (**target = 0**) based on customer history, financial metrics, shipping lead times, and product risk profiles.

2. Dataset Overview & Schema

The analysis is built upon **1,000 historical order records** containing 6 predictor variables and 1 binary target label. Data validation confirmed 0 missing values, 0 duplicate records, and a perfectly balanced 50/50 target distribution.

Feature	Type	Min	Mean	Max	Description
order_value	Float	\$11.90	\$190.19	\$2659.39	Total monetary value of the order
item_count	Integer	0	2.98	9	Number of individual items in order
discount_pct	Float	0.00%	50.43%	100.00%	Discount percentage granted
customer_return_rate	Float	0.00%	50.75%	100.00%	Historical return rate of customer
delivery_days	Integer	0 days	29.71 days	60 days	Delivery lead time in days
category_risk	Integer	1	3.00	5	Product category risk tier (1-5)
target	Binary	0	0.50	1	Order outcome (1 = Return, 0 = Keep)

3. Model Performance Benchmark

A baseline **Logistic Regression** model was trained on 80% (800 samples) with StandardScaler feature normalization and evaluated on a stratified 20% test set (200 samples).

67.00%	65.45%	72.00%	68.57%	73.07%
ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC

Test Set Confusion Matrix (N = 200)

	Predicted: No Return (0)	Predicted: Return (1)
Actual: No Return (0)	62 (True Negative)	38 (False Positive)
Actual: Return (1)	28 (False Negative)	72 (True Positive)

4. Feature Importance & Return Drivers

Standardized model coefficients illustrate the relative weight and direction of each predictor in determining return probability:

Feature Name	Coefficient	Impact Direction	Analytical Takeaway
delivery_days	+0.6429	Strong Positive	Shipping latency is the single largest risk factor.
category_risk	-0.6069	Negative	Category risk tier baseline linear coefficient.
discount_pct	+0.5303	Moderate Positive	Heavy discounts incentivize impulse buys.
order_value	+0.4945	Moderate Positive	High value items undergo stricter post-delivery scrutiny.
item_count	-0.4312	Negative	Larger basket size lowers return risk slightly.
customer_return_rate	-0.3455	Slight Negative	Requires non-linear modeling & domain review.

5. Strategic Recommendations & Roadmap

1. Fast-Track Shipping for High Risk Orders: Since delivery_days (+0.6429) is the largest risk driver, expediting dispatch or offering priority shipping for orders exceeding \$150 or 40% discount directly cuts return probability.
2. Targeted Discount Rules: High discount orders (+0.5303) show elevated return propensity. Consider implementing final-sale policies or proactive post-purchase onboarding messages for items discounted > 50%.
3. Model Expansion (XGBoost / LightGBM): Transition to tree-based ensemble models to capture non-linear relationships, and tune classification decision thresholds based on actual reverse logistics cost matrices.